Robert Barthell Garay

corigarayv@gmail.com https://www.linkedin.com/in/robert-barthell-garay/ robertbarthellgaray.github.io

EDUCATION

University of Illinois Urbana-Champaign — Expected December 2026
Bachelor of Science in Aerospace Engineering — 3.3 GPA

SKILLS

Technical Skills: FreeFlyer (trajectory propagation, perturbation modeling, ADCS), Python (numerical optimization, adaptive weighting), Blender (3D Modeling, asset development, data analysis, data visualization), Unreal Engine 5 (3D Rendering/Animation)
Coursework: AE 502 (Advanced Orbital Mechanics), AE 442/443 (Spacecraft Systems Design), ENG 491 (Cubesats)
Languages: English, Spanish (native), Mandarin Chinese (HSK5), Russian, German, French, Portuguese, Hindi, Fusha Arabic

WORK EXPERIENCE:

University of Illinois - AE 402 Course Assistant — **2026 Aug - Present**

- Produced and edited hyperrealistic spacecraft trajectory render multimedia to supplement class presentations and teach history
- Assembled pipeline to convert and import physically accurate orbit data (Python/JPL Horizons) into 3D Rendering software
- Developed and maintained orbital mechanics assignments with in-depth explanations and historical scenarios, expanding breadth
- Hosted office hours, aided students with problem solving and conceptual understanding of orbits, adding new depth to class

National Aeronautics and Space Administration - Software Development Internship — **2024 Jun - 2024 Aug**

- Developed high-fidelity interactive training content and environment enhancements for NASA simulator of Lunar/Martian surface
- Created 3D models of EVA tools, habitats, and equipment which will be used for VR training of future astronaut candidates
- Collaborated with software engineers to implement interactive physics-based systems and scenarios of extravehicular activities

RESEARCH/ORGANIZATIONAL EXPERIENCE:

Laboratory for Advanced Space Systems at Illinois - Research Assistant — **2025 Aug - Present**

CubeSail II - Feasibility mission testing deployment and controllability of Solar Sail propulsion

- Coordinating the mission's early concept development stages, literature review, created initial simulation framework
- Refined simulated attitude determination logic for sun-pointing control to maximize solar exposure and efficiency / orbit raising
- Adapted orbital integrator to simulate flexible spacecraft structures and load tension under solar radiation pressure
- Created multi-node dynamics soft-body/tether system to model structural deformation in sun synchronous terminator orbit

MonARCH - CubeSat testing a novel electro spray-based hybrid propulsion system

- Programmed long-term orbit propagation simulations producing data about propulsion efficiency and impact of external forces
- Modeled coupled ADCS-orbit dynamics accounting for momentum build-up and extended propulsion-induced disturbances
- Built and implemented custom models for counteracting misaligned thrust and maintaining attitude stability using torque rods
- Achieved sustained reaction wheel momentum containment using optimized torque rod desaturation strategy

Chinese Language Association for International Development - Board Member — **2024 Jan - 2026 Oct**

- Led discussion and conversation tables for Chinese language learners, produced culturally immersive learning/vocabulary content

Illinois Space Society - Media Team, 3D Animator, Undergraduate Design Member — **2023 Sep - 2025 Apr**

- Analyzed link budget capacity of an orbital laser satellite communications network around Moon using FreeFlyer
- Digitally reconstructed the Lunar South Pole in Unreal Engine and created 3D renders and animations of proposal designs
- Produced video recaps/ written reports of rocket launch tests, directed advertising shown to hundreds of prospective members

PROJECT EXPERIENCE:

Robert's Lambert Multi Gravity Assist Solver (RoLaMGAS) — **2026 Oct - Present**

- Created and published a tool to find MGA trajectories for interplanetary spacecraft ensuring adherence to C3, fuel and time limits
- Reconstructed historical interplanetary spacecraft trajectories with tool, ensured accurate results within real mission timeframes
- Authored recreations of active spacecraft proposals in astronomy software Celestia using data integrated by solver

Spacecraft Systems Design Proposal — **2026 Jan - 2026 May**

- Co-wrote research paper proposing twin spacecraft solution for AIAA Solar Weather Monitoring design competition
- Designed trajectories for Mars-Sun L1 and gravity assists for 30* inclined relay probe, ensuring constant Earth-Mars line of sight

Advanced Orbital Mechanics Semester Project — **2026 Feb - 2026 Apr**

- Constructed n-body propagator to design multiple gravity-assist trajectory (EΔvEJU) for Uranus orbiter mission
- Implemented shooting method with least squares solver which achieved 40% reduction in needed Δv for probe
- Expanded concepts from published NASA mission studies, gained experience with nonlinear trajectory optimization